CADCAST Cost Model — Notation

Convention: If a parameter repeats across sections, it keeps the same symbol (e.g., N_{parts}, $t_{\text{build,bj}}$).

Binder Jetting Sand Casting

Material Cost per Part

Material cost per part in binder jetting sand casting includes two main components:

- the cost of consumables used in the printing process (sand powder and binder), and
- the direct cost of the metal to be poured during sand casting.

The formulation distributes the consumable cost across the batch and adds the calculated metal cost considering casting density, volume, and the metal-loss factor.

$$C^{\text{mat}}_{/\text{part}} = \frac{m_{\text{sand}}\, r_{\text{sand}} + V_{\text{binder}}\, r_{\text{binder}}}{B_{\text{batch}}} + C_{\text{metal}} \tag{BJ-1}$$

$$m_{\text{sand}} = \rho_{\text{sand}}\, k_{\text{box}}\, 1.5\, B_{\text{batch}}\, V_{\text{box,part}} \tag{BJ-2}$$

$$V_{\text{binder}} = V_{\text{print}}\,(1-\phi_{\text{pack}})\, S_{\text{binder}}\, f_{\text{ineff}} \times 10^{-6} \tag{BJ-3}$$

$$C_{\text{metal}} = r_{\text{metal}}\, m_{\text{cast}}\, f_{\text{loss}} \tag{BJ-4}$$

$$m_{\text{cast}} = \rho_{\text{metal}}\, V_{\text{cast}} \tag{BJ-5}$$

where:

$\boldsymbol{C^{\text{mat}}_{/\text{part}}}$ material cost per part [$]

$\boldsymbol{m_{\text{sand}}}$ total sand used per batch [kg] (see (BJ-2))

$\boldsymbol{r_{\text{sand}}}$ powder/sand price [$/kg]

$\boldsymbol{V_{\text{binder}}}$ total binder used per batch [L]

$\boldsymbol{V_{\text{print}}}$ printed region volume from STL [mm^3]

$\boldsymbol{\phi_{\text{pack}}}$ packing rate (fraction of packed sand) [–]

$\boldsymbol{S_{\text{binder}}}$ binder saturation [–]

$\boldsymbol{f_{\text{ineff}}}$ binder inefficiency multiplier [–]

$\boldsymbol{r_{\text{binder}}}$ binder price [$/L]

$B_{\mathbf{batch}}$ batch size [–]

$r_{\mathbf{metal}}$ unit metal cost [$/kg]

$\rho_{\mathbf{metal}}$ casting metal density [kg/m^3]

$V_{\mathbf{cast}}$ net casting volume per part [m^3]

$m_{\mathbf{cast}}$ casting weight (net metal) per part [kg]

$f_{\mathbf{loss}}$ factor for metal loss [–]

$C_{\mathbf{metal}}$ direct cost of metal per part [$]

$\rho_{\mathbf{sand}}$ sand density [kg/m^3]

$k_{\mathbf{box}}$ mold bounding-box factor (accounts for flask, riser, gating, etc.) [–]

$V_{\mathbf{box,part}}$ bounding-box volume of one part [m^3]

Machine Cost per Part

Machine cost per part in binder jetting sand casting is composed of two contributions:

- the cost of printer usage, determined by distributing the machine's investment cost (amortized over its lifetime and utilization) across the available machine hours; and
- the cost of furnace usage, determined from the furnace hourly rate and distributed across the batch.

Together, these two contributions define the per-part machine cost.

$$C^{\text{mach}}_{/\text{part}} = \frac{C^{/h}_{\text{mach,bj}}\, t_{\text{build,bj}}}{B_{\text{batch}}} + \frac{C^{/h}_{\text{furn}}\, t_{\text{firing}}}{B_{\text{batch}}} \tag{BJ-6}$$

$$C^{/h}_{\text{mach,bj}} = \frac{C_{\text{inv,bj}}}{\eta_{\text{mach,bj}}\, H_{\text{year,bj}}} \tag{BJ-7}$$

$$C_{\text{inv,bj}} = \frac{C_{\text{mach,bj}}}{L_{\text{mach,bj}}} \tag{BJ-8}$$

$$C^{/h}_{\text{furn}} = \frac{C_{\text{inv,sand}}}{\eta_{\text{furn}}\, H_{\text{year,sand}}} \tag{BJ-9}$$

$$C_{\text{inv,sand}} = \frac{C_{\text{mach,sand}}}{L_{\text{mach,sand}}} \tag{BJ-10}$$

where:

$C^{\textbf{mach}}_{\textbf{/part}}$ machine cost per part [$]

$C^{/h}_{\textbf{mach,bj}}$ machine hourly cost (BJ printer) [$/h]

$t_{\textbf{build,bj}}$ binder jetting build time [h]

$B_{\textbf{batch}}$ batch size [–]

$C^{/h}_{\textbf{furn}}$ furnace hourly rate [$/h]

$t_{\textbf{firing}}$ furnace firing time [h]

$C_{\textbf{inv,bj}}$ BJ printer investment cost [$/year]

$\eta_{\textbf{mach,bj}}$ BJ printer utilization [–]

$H_{\textbf{year,bj}}$ total BJ printer machine hours per year [h/year]

$C_{\textbf{mach,bj}}$ BJ printer acquisition cost [$]

$L_{\mathbf{mach,bj}}$ BJ printer lifetime [years]

$C_{\mathbf{inv,sand}}$ sand-casting furnace investment cost [$/year]

$C_{\mathbf{mach,sand}}$ furnace acquisition cost (sand casting) [$]

$L_{\mathbf{mach,sand}}$ furnace lifetime [years]

$\eta_{\mathbf{furn}}$ furnace utilization [–]

$H_{\mathbf{year,sand}}$ total furnace hours per year [h/year]

Labor Cost per Part

Labor cost per part in binder jetting sand casting has two components:

- **Binder jetting** total operator time (single labor rate), and
- **Sand-casting** technical labor (technician man-hours, with rejection).

These contributions are combined and divided across the batch to obtain the labor cost per part.

$$C^{\text{lab}}_{/\text{part}} = \frac{r_{\text{lab}}\, t_{\text{op}} + C_{\text{techlab}}}{B_{\text{batch}}} \tag{BJ-11}$$

$$C_{\text{techlab}} = f_{\text{reject}}\, r_{\text{tech}}\, H_{\text{tech,sand}} \tag{BJ-12}$$

where:

$C^{\mathbf{lab}}_{\mathbf{/part}}$ labor cost per part [$]

$B_{\mathbf{batch}}$ batch size [–]

$r_{\mathbf{lab}}$ labor rate for binder jetting activities [$/h]

$t_{\mathbf{op}}$ binder jetting total operator time [h]

$C_{\mathbf{techlab}}$ sand-casting technical labor subtotal [$]

$f_{\mathbf{reject}}$ activity rejection factor (sand casting) [–]

$r_{\mathbf{tech}}$ technician hourly rate (sand-casting activities) [$/h]

$H_{\mathbf{tech,sand}}$ sand-casting technician man-hours [h]

Energy Cost per Part

Energy cost per part in binder jetting sand casting comprises:

- the printer's electrical energy during the BJ build (allocated over the batch), and
- the foundry energy required to melt and hold the casting metal prior to pouring.

$$C^{\text{eng}}_{/\text{part}} = \frac{r_{\text{elec}}\, P_{\text{print,bj}}\, t_{\text{build,bj}}}{B_{\text{batch}}} + C_{\text{melt}} + C_{\text{hold}} \tag{BJ-13}$$

$$C_{\text{melt}} = r_{\text{elec}} \frac{E_{\text{melt}}}{\eta_{\text{furn}}}\, f_{\text{melt}}\, \rho_{\text{metal}}\, V_{\text{part}} \times 10^{-9} \tag{BJ-14}$$

$$C_{\text{hold}} = r_{\text{elec}} \frac{E_{\text{hold}}}{\eta_{\text{furn}}}\, f_{\text{melt}}\, \rho_{\text{metal}}\, V_{\text{part}}\, t_{\text{hold}} \times 10^{-9} \tag{BJ-15}$$

where:

$C^{\text{eng}}_{/\text{part}}$ energy cost per part [$]

r_{elec} electricity rate [$/kWh]

$P_{\text{print,bj}}$ printer average electrical power (BJ) [kW]

$t_{\text{build,bj}}$ BJ build time [h]

B_{batch} batch size [–]

C_{melt} melting energy cost per part [$]

E_{melt} specific energy to melt metal [kWh/kg]

C_{hold} holding energy cost per part [$]

E_{hold} specific energy to hold molten metal [kWh/(kg·h)]

t_{hold} holding time [h]

η_{furn} furnace/thermal efficiency (0–1) [–]

f_{melt} melt allowance factor for gating/runners/scrap (e.g., 1.2) [–]

ρ_{metal} casting metal density [kg/m^3]

V_{part} net part volume [mm^3]

Note: For typical binder jetting sand casting, multiple identical molds or cores are poured per melt batch. The melt and hold energy terms are therefore independent of batch size for identical parts; only the BJ printer term scales with $1/B_{\text{batch}}$.

Post-Processing Cost per Part

Post-processing cost per part is modeled as:

- a percentage factor representing finishing operations (e.g., breakout, blast/cleaning), applied to
- the direct cost per part excluding labor (i.e., material + machine + energy).

$$C^{\text{post}}_{/\text{part}} = f_{\text{post}}\, C^{\text{direct}\backslash\text{lab}}_{/\text{part}} \tag{BJ-16}$$

$$C^{\text{direct}\backslash\text{lab}}_{/\text{part}} = C^{\text{mat}}_{/\text{part}} + C^{\text{mach}}_{/\text{part}} + C^{\text{eng}}_{/\text{part}} \tag{BJ-17}$$

where:

$\boldsymbol{C^{\textbf{post}}_{/\textbf{part}}}$ post-processing cost per part [$]

$\boldsymbol{f_{\textbf{post}}}$ post-processing factor (dimensionless; e.g., 0.10 = 10%) [–]

$\boldsymbol{C^{\textbf{direct}\backslash\textbf{lab}}_{/\textbf{part}}}$ direct cost per part excluding labor [$]

$\boldsymbol{C^{\textbf{mat}}_{/\textbf{part}}}$ material cost per part [$]

$\boldsymbol{C^{\textbf{mach}}_{/\textbf{part}}}$ machine cost per part [$]

$\boldsymbol{C^{\textbf{eng}}_{/\textbf{part}}}$ energy cost per part [$]

Maintenance Cost per Part

Maintenance cost per part in binder jetting sand casting is modeled as:

- a yearly maintenance expense, expressed as a percentage of the printer acquisition cost, and
- an allocation of this yearly expense across the machine's available hours and the batch size.

$$C_{\text{mnt}}^{/year} = \alpha_{\text{mnt}}\, C_{\text{mach,bj}} \tag{BJ-18}$$

$$C_{/\text{part}}^{\text{mnt}} = \frac{C_{\text{mnt}}^{/year}}{\eta_{\text{mach,bj}}\, H_{\text{year,bj}}}\, \frac{t_{\text{build,bj}}}{B_{\text{batch}}} \tag{BJ-19}$$

where:

$C_{\mathbf{mnt}}^{/year}$ yearly maintenance cost [$/year]

$\alpha_{\mathbf{mnt}}$ maintenance fraction of printer acquisition cost (per year) [–]

$C_{\mathbf{mach,bj}}$ printer acquisition cost [$]

$C_{\mathbf{/part}}^{\mathbf{mnt}}$ maintenance cost per part [$]

$\eta_{\mathbf{mach,bj}}$ printer utilization [–]

$H_{\mathbf{year,bj}}$ total printer operating hours per year [h/year]

$t_{\mathbf{build,bj}}$ binder jetting build time [h]

$B_{\mathbf{batch}}$ batch size [–]

Overhead Cost per Part

Overhead cost per part is modeled separately for:

- the binder jetting process, based on the annual overhead for the printer, and
- the sand casting process, based on the annual overhead for foundry operations.

In both cases, the annual overhead cost is expressed as a percentage of the machine acquisition cost and then distributed across yearly machine hours (adjusted by utilization) and the *batch size.*

$$C_{\mathrm{oh,bj}}^{/year} = \beta_{\mathrm{oh,bj}}\, C_{\mathrm{mach,bj}} \tag{BJ-20}$$

$$C_{\mathrm{oh,bj}}^{/h} = \frac{C_{\mathrm{oh,bj}}^{/year}}{\eta_{\mathrm{mach,bj}}\, H_{\mathrm{year,bj}}} \tag{BJ-21}$$

$$C_{/\mathrm{part}}^{\mathrm{oh,bj}} = \frac{C_{\mathrm{oh,bj}}^{/h}\, t_{\mathrm{build,bj}}}{B_{\mathrm{batch}}} \tag{BJ-22}$$

$$C_{\mathrm{oh,sand}}^{/year} = \beta_{\mathrm{oh,sand}}\, C_{\mathrm{mach,sand}} \tag{BJ-23}$$

$$C_{\mathrm{oh,sand}}^{/h} = \frac{C_{\mathrm{oh,sand}}^{/year}}{\eta_{\mathrm{mach,sand}}\, H_{\mathrm{year,sand}}} \tag{BJ-24}$$

$$C_{/\mathrm{part}}^{\mathrm{oh,sand}} = \frac{C_{\mathrm{oh,sand}}^{/h}\, t_{\mathrm{batch}}}{B_{\mathrm{batch}}} \tag{BJ-25}$$

$$C_{/\mathrm{part}}^{\mathrm{oh,total}} = C_{/\mathrm{part}}^{\mathrm{oh,bj}} + C_{/\mathrm{part}}^{\mathrm{oh,sand}} \tag{BJ-26}$$

where:

$C_{\mathbf{oh,bj}}^{/year}$ annual overhead cost (BJ) [$/year]

$\beta_{\mathbf{oh,bj}}$ overhead percentage of acquisition (BJ) [–]

$C_{\mathbf{mach,bj}}$ machine acquisition cost (BJ) [$]

$C_{\mathbf{oh,bj}}^{/h}$ overhead rate (BJ) [$/h]

$\eta_{\mathbf{mach,bj}}$ machine utilization (BJ) [–]

$H_{\mathbf{year,bj}}$ annual machine hours (BJ) [h/year]

$C_{\mathbf{/part}}^{\mathbf{oh,bj}}$ overhead cost per part (BJ) [$]

$t_{\mathbf{build,bj}}$ build time [h]

$C^{/year}_{\mathbf{oh,sand}}$ annual overhead cost (sand casting) [\$/year]

$\beta_{\mathbf{oh,sand}}$ overhead percentage of acquisition (sand casting) [–]

$C_{\mathbf{mach,sand}}$ machine acquisition cost (sand casting) [\$]

$C^{/h}_{\mathbf{oh,sand}}$ overhead rate (sand casting) [\$/h]

$\eta_{\mathbf{mach,sand}}$ machine utilization (sand casting) [–]

$H_{\mathbf{year,sand}}$ annual machine hours (sand casting) [h/year]

$C^{\mathbf{oh,sand}}_{\mathbf{/part}}$ overhead cost per part (sand casting) [\$]

$t_{\mathbf{batch}}$ batch time (sand casting) [h]

$C^{\mathbf{oh,total}}_{\mathbf{/part}}$ total overhead cost per part [\$]

$B_{\mathbf{batch}}$ batch size [–]

Total Cost per Part

The total cost per part is the sum of:

- direct material cost (BJ consumables + poured metal),
- machine cost (printer + furnace components),
- labor cost (preprocessing, operator, engineer, technical),
- energy cost (BJ electrical + melting and holding),
- post-processing cost,
- maintenance cost, and
- total overhead cost.

$$C^{\text{total}}_{/\text{part}} = C^{\text{mat}}_{/\text{part}} + C^{\text{mach}}_{/\text{part}} + C^{\text{lab}}_{/\text{part}} + C^{\text{eng}}_{/\text{part}} + C^{\text{post}}_{/\text{part}} + C^{\text{mnt}}_{/\text{part}} + C^{\text{oh,total}}_{/\text{part}} \tag{BJ-27}$$

where:

$C^{\mathbf{total}}_{\mathbf{/part}}$ total cost per part [$]

$C^{\mathbf{mat}}_{\mathbf{/part}}$ material cost per part [$]

$C^{\mathbf{mach}}_{\mathbf{/part}}$ machine cost per part [$]

$C^{\mathbf{lab}}_{\mathbf{/part}}$ labor cost per part [$]

$C^{\mathbf{eng}}_{\mathbf{/part}}$ energy cost per part [$]

$C^{\mathbf{post}}_{\mathbf{/part}}$ post-processing cost per part [$]

$C^{\mathbf{mnt}}_{\mathbf{/part}}$ maintenance cost per part [$]

$C^{\mathbf{oh,total}}_{\mathbf{/part}}$ overhead cost per part [$]

DLP Ceramic Investment Casting

Material Cost per Part

Material cost per part in DLP ceramic investment casting includes:

- the resin used to print the mold, allocated over the batch size; and
- the casting metal, taken as the per-part gross metal (net mass scaled by melt allowance) times the unit metal price plus loss surcharge.

$$C^{\mathrm{mat}}_{/\mathrm{part}} = r_{\mathrm{resin}}\,\frac{V_{\mathrm{resin}}}{1000}\,\frac{1}{B_{\mathrm{batch}}} + \left(r_{\mathrm{mat}} + r_{\mathrm{loss}}\right) f_{\mathrm{melt}}\, m_{\mathrm{part}} \tag{DLP-1}$$

$$m_{\mathrm{part}} = \rho_{\mathrm{metal}}\, V_{\mathrm{part}} \times 10^{-9} \tag{DLP-2}$$

where:

$C^{\mathbf{mat}}_{\mathbf{/part}}$ material cost per part [$]

$r_{\mathbf{resin}}$ resin price [$/mL]

$V_{\mathbf{resin}}$ resin volume to print one mold [mm^3] (mm$^3 \to$ mL: divide by 1000)

$B_{\mathbf{batch}}$ Batch size [–]

$r_{\mathbf{mat}}$ raw casting metal price [$/kg]

$r_{\mathbf{loss}}$ process-loss surcharge per kg of *gross* metal [$/kg]

$f_{\mathbf{melt}}$ melt allowance factor (e.g., 1.2) [–]

$m_{\mathbf{part}}$ net mass of one finished part [kg]

$\rho_{\mathbf{metal}}$ casting metal density [kg/m^3]

$V_{\mathbf{part}}$ net part volume [mm^3]

Note: For typical DLP investment casting, $V_{\mathrm{resin}} \neq V_{\mathrm{part}}$.

Machine Cost per Part

Machine cost per part in DLP ceramic investment casting is composed of two components:

- the cost of printer usage, determined by distributing the machine's investment cost (amortized over its lifetime and utilization) across the available machine hours; and
- the cost of furnace usage, determined from the furnace hourly rate and distributed across the batch size.

$$C^{\mathrm{mach}}_{/\mathrm{part}} = \frac{C^{/h}_{\mathrm{mach,dlp}}\, t_{\mathrm{build,dlp}}}{B_{\mathrm{batch}}} + \frac{C^{/h}_{\mathrm{furn}}\, t_{\mathrm{firing}}}{B_{\mathrm{batch}}} \tag{DLP-3}$$

$$C^{/h}_{\mathrm{mach,dlp}} = \frac{C_{\mathrm{inv,dlp}}}{\eta_{\mathrm{mach,dlp}}\, H_{\mathrm{year,dlp}}} \tag{DLP-4}$$

$$C_{\mathrm{inv,dlp}} = \frac{C_{\mathrm{mach,dlp}}}{L_{\mathrm{mach,dlp}}} \tag{DLP-5}$$

$$C^{/h}_{\mathrm{furn}} = \frac{C_{\mathrm{inv,furn}}}{\eta_{\mathrm{furn}}\, H_{\mathrm{year,furn}}} \tag{DLP-6}$$

$$C_{\mathrm{inv,furn}} = \frac{C_{\mathrm{mach,furn}}}{L_{\mathrm{mach,furn}}} \tag{DLP-7}$$

where:

$C^{\mathbf{mach}}_{\mathbf{/part}}$ machine cost per part [$]

$C^{/h}_{\mathbf{mach,dlp}}$ DLP printer hourly cost [$/h]

$t_{\mathbf{build,dlp}}$ DLP build time [h]

$B_{\mathbf{batch}}$ Batch size [–]

$C^{/h}_{\mathbf{furn}}$ furnace hourly rate [$/h]

$t_{\mathbf{firing}}$ furnace firing time [h]

$C_{\mathbf{inv,dlp}}$ DLP printer investment cost [$/year]

$\eta_{\mathbf{mach,dlp}}$ DLP printer utilization [–]

$H_{\mathbf{year,dlp}}$ total DLP printer machine hours per year [h/year]

$C_{\mathbf{mach,dlp}}$ DLP printer acquisition cost [$]

$L_{\mathbf{mach,dlp}}$ DLP printer lifetime [years]

$C_{\mathbf{inv,furn}}$ furnace investment cost [$/year]

$C_{\mathbf{mach,furn}}$ furnace acquisition cost [$]

$L_{\mathbf{mach,furn}}$ furnace lifetime [years]

$\eta_{\mathbf{furn}}$ furnace utilization [–]

$H_{\mathbf{year,furn}}$ total furnace hours per year [h/year]

Labor Cost per Part

Labor cost per part includes:

- total operator time for DLP activities (preprocessing, support removal, post-cure, drying), and
- breakout and blast cleaning costs allocated over the batch size.

$$C^{\text{lab}}_{/\text{part}} = \frac{r_{\text{lab}}\, t_{\text{op}}}{B_{\text{batch}}} + \frac{C_{\text{break}} + C_{\text{blast}}}{B_{\text{batch}}} \tag{DLP-8}$$

where:

$C^{\mathbf{lab}}_{\mathbf{/part}}$ labor cost per part [$]

$r_{\mathbf{lab}}$ labor rate [$/h]

$t_{\mathbf{op}}$ total operator time per batch (DLP) [h]

$C_{\mathbf{break}}$ breakout cost per mold [$]

$C_{\mathbf{blast}}$ blast cleaning cost per mold [$]

$B_{\mathbf{batch}}$ batch size [–]

Energy Cost per Part

Energy cost per part in DLP ceramic investment casting comprises:

- the printer's electrical energy during the DLP build, and
- the foundry energy to melt the casting metal (per part, adjusted by furnace efficiency).

$$C^{\text{eng}}_{/\text{part}} = \frac{r_{\text{elec}}\, P_{\text{print,dlp}}\, t_{\text{build,dlp}}}{B_{\text{batch}}} + C_{\text{melt}} \tag{DLP-9}$$

$$C_{\text{melt}} = r_{\text{elec}}\, \frac{E_{\text{melt}}}{\eta_{\text{furn}}}\, f_{\text{melt}}\, \rho_{\text{metal}}\, V_{\text{part}} \times 10^{-9} \tag{DLP-10}$$

where:

$C^{\textbf{eng}}_{\textbf{/part}}$ energy cost per part [$]

$r_{\textbf{elec}}$ electricity rate [$/kWh]

$P_{\textbf{print,dlp}}$ printer average electrical power (DLP) [kW]

$t_{\textbf{build,dlp}}$ DLP build time [h]

$B_{\textbf{batch}}$ Batch size [–]

$C_{\textbf{melt}}$ melting energy cost (per part) [$]

$E_{\textbf{melt}}$ specific energy to melt metal [kWh/kg]

$\eta_{\textbf{furn}}$ furnace/thermal efficiency (0–1) [–]

$f_{\textbf{melt}}$ melt allowance factor for gating/runners/sprue (e.g., 1.2) [–]

$\rho_{\textbf{metal}}$ casting metal density [kg/m^3]

$V_{\textbf{part}}$ net part volume [mm^3]

Note: For typical DLP investment casting, molds contain multiple identical parts; thus the per-part melt energy cost is independent of batch size. If a mixed-geometry mold is used, melt energy may be allocated per part proportional to its volume V_i.

Post-Processing Cost per Part

Post-processing cost per part in DLP ceramic investment casting is modeled as:

- a percentage factor representing finishing operations (e.g., breakout, blast/cleaning), applied to
- the direct cost per part excluding labor (i.e., material + machine + energy).

$$C^{\text{post}}_{/\text{part}} = f_{\text{post}}\, C^{\text{direct}\backslash\text{lab}}_{/\text{part}} \tag{DLP-11}$$

$$C^{\text{direct}\backslash\text{lab}}_{/\text{part}} = C^{\text{mat}}_{/\text{part}} + C^{\text{mach}}_{/\text{part}} + C^{\text{eng}}_{/\text{part}} \tag{DLP-12}$$

where:

$C^{\mathbf{post}}_{\mathbf{/part}}$ post-processing cost per part [$]

$f_{\mathbf{post}}$ post-processing factor [–]

$C^{\mathbf{direct}\backslash\mathbf{lab}}_{\mathbf{/part}}$ direct cost per part excluding labor [$]

$C^{\mathbf{mat}}_{\mathbf{/part}}$ material cost per part [$]

$C^{\mathbf{mach}}_{\mathbf{/part}}$ machine cost per part [$]

$C^{\mathbf{eng}}_{\mathbf{/part}}$ energy cost per part [$]

Maintenance Cost per Part

Maintenance cost per part in DLP ceramic investment casting is modeled as:

- a yearly maintenance expense, expressed as a percentage of the printer acquisition cost, and
- allocation of this yearly expense across available machine hours and the batch size.

$$C_{\text{mnt}}^{/year} = \alpha_{\text{mnt}}\, C_{\text{mach,dlp}} \tag{DLP-13}$$

$$C_{/\text{part}}^{\text{mnt}} = \frac{C_{\text{mnt}}^{/year}}{\eta_{\text{mach,dlp}}\, H_{\text{year,dlp}}}\, \frac{t_{\text{build,dlp}}}{B_{\text{batch}}} \tag{DLP-14}$$

where:

$C_{\mathbf{mnt}}^{/year}$ yearly maintenance cost [$/year]

$\alpha_{\mathbf{mnt}}$ maintenance fraction of printer acquisition cost (per year) [–]

$C_{\mathbf{mach,dlp}}$ DLP printer acquisition cost [$]

$C_{\mathbf{/part}}^{\mathbf{mnt}}$ maintenance cost per part [$]

$\eta_{\mathbf{mach,dlp}}$ DLP printer utilization [–]

$H_{\mathbf{year,dlp}}$ total DLP printer operating hours per year [h/year]

$t_{\mathbf{build,dlp}}$ DLP build time [h]

$B_{\mathbf{batch}}$ Batch size [–]

Overhead Cost per Part

Overhead cost per part in DLP ceramic investment casting is modeled as:

- a yearly overhead expense, expressed as a percentage of the printer acquisition cost, and
- allocation of this yearly overhead across available machine hours and the batch size.

$$C_{\mathrm{oh,dlp}}^{/year} = \beta_{\mathrm{oh,dlp}}\, C_{\mathrm{mach,dlp}} \tag{DLP-15}$$

$$C_{\mathrm{oh,dlp}}^{/h} = \frac{C_{\mathrm{oh,dlp}}^{/year}}{\eta_{\mathrm{mach,dlp}}\, H_{\mathrm{year,dlp}}} \tag{DLP-16}$$

$$C_{/\mathrm{part}}^{\mathrm{oh}} = \frac{C_{\mathrm{oh,dlp}}^{/h}\, t_{\mathrm{build,dlp}}}{B_{\mathrm{batch}}} \tag{DLP-17}$$

where:

$C_{\mathbf{oh,dlp}}^{/year}$ annual overhead cost (DLP) [$/year]

$\beta_{\mathbf{oh,dlp}}$ overhead fraction of acquisition (per year) [–]

$C_{\mathbf{mach,dlp}}$ DLP printer acquisition cost [$]

$C_{\mathbf{oh,dlp}}^{/h}$ overhead rate [$/h]

$\eta_{\mathbf{mach,dlp}}$ DLP printer utilization [–]

$H_{\mathbf{year,dlp}}$ total DLP printer operating hours per year [h/year]

$t_{\mathbf{build,dlp}}$ DLP build time [h]

$C_{\mathbf{/part}}^{\mathbf{oh}}$ overhead cost per part [$]

$B_{\mathbf{batch}}$ Batch size [–]

Total Cost per Part

The total cost per part in DLP ceramic investment casting is the sum of:

- direct material cost (resin and casting metal),
- machine cost (printer and furnace),
- labor cost (preprocessing, support removal, post-curing, drying, breakout, blasting),
- energy cost (printing and melting),
- post-processing cost,
- maintenance cost, and
- overhead cost.

$$C^{\text{total}}_{/\text{part}} = C^{\text{mat}}_{/\text{part}} + C^{\text{mach}}_{/\text{part}} + C^{\text{lab}}_{/\text{part}} + C^{\text{eng}}_{/\text{part}} + C^{\text{post}}_{/\text{part}} + C^{\text{mnt}}_{/\text{part}} + C^{\text{oh}}_{/\text{part}} \tag{DLP-18}$$

where:

$C^{\textbf{total}}_{\textbf{/part}}$ total cost per part [$]

$C^{\textbf{mat}}_{\textbf{/part}}$ material cost per part [$]

$C^{\textbf{mach}}_{\textbf{/part}}$ machine cost per part [$]

$C^{\textbf{lab}}_{\textbf{/part}}$ labor cost per part [$]

$C^{\textbf{eng}}_{\textbf{/part}}$ energy cost per part [$]

$C^{\textbf{post}}_{\textbf{/part}}$ post-processing cost per part [$]

$C^{\textbf{mnt}}_{\textbf{/part}}$ maintenance cost per part [$]

$C^{\textbf{oh}}_{\textbf{/part}}$ overhead cost per part [$]

Printed Pattern Investment Casting (Lost-PLA)

Material Cost per Part

Material cost per part in printed pattern (Lost-PLA) investment casting includes:

- the polymer filament used to print the pattern (including supports),
- the ceramic slurry used for shell building (allocated over the batch), and
- the casting metal, expressed per part using geometry and melt allowance.

$$C^{\mathrm{mat}}_{/\mathrm{part}} = r_{\mathrm{fil}}\,\rho_{\mathrm{fil}}\,(V_{\mathrm{part}} + V_{\mathrm{support}}) \times 10^{-9} + \frac{C_{\mathrm{slurry,batch}}}{B_{\mathrm{batch}}} + C_{\mathrm{metal}} \tag{PPI-1}$$

$$C_{\mathrm{metal}} = (r_{\mathrm{mat}} + r_{\mathrm{loss}})\, f_{\mathrm{melt}}\,\rho_{\mathrm{metal}}\,V_{\mathrm{part}} \times 10^{-9} \tag{PPI-2}$$

$$C_{\mathrm{slurry,batch}} = r_{\mathrm{slurry}}\, m_{\mathrm{slurry,batch}} \tag{PPI-3}$$

$$m_{\mathrm{slurry,batch}} = \big(A_{\mathrm{tree,part}}\,B_{\mathrm{batch}} + A_{\mathrm{sprue,batch}}\big)\, t_{\mathrm{shell}}\,\rho_{\mathrm{cer}}\,\phi_{\mathrm{sol}} \tag{PPI-4}$$

$$A_{\mathrm{sprue,batch}} = \gamma_{\mathrm{sprue}}\; B_{\mathrm{batch}}\; A_{\mathrm{tree,part}} \tag{PPI-5}$$

where:

$C^{\mathbf{mat}}_{\mathbf{/part}}$ material cost per part [$/part]

$r_{\mathbf{fil}}$ FDM filament price [$/kg]

$\rho_{\mathbf{fil}}$ filament density [kg/m^3]

$V_{\mathbf{part}}$ net part volume (from STL) [mm^3]

$V_{\mathbf{support}}$ support volume for the printed pattern [mm^3]

$B_{\mathbf{batch}}$ batch size [–]

$C_{\mathbf{metal}}$ casting metal cost per part [$]

$r_{\mathbf{mat}}$ raw casting metal price [$/kg]

$r_{\mathbf{loss}}$ process-loss surcharge per kg of *gross* metal [$/kg]

$f_{\mathbf{melt}}$ melt allowance factor (e.g., 1.2) [–]

$\rho_{\mathbf{metal}}$ casting metal density [kg/m^3]

$r_{\mathbf{slurry}}$ ceramic slurry price [$/kg]

$C_{\textbf{slurry,batch}}$ slurry cost per batch [$]

$m_{\textbf{slurry,batch}}$ slurry mass per batch [kg]

$A_{\textbf{tree,part}}$ pattern-tree surface area per part [m^2]

$A_{\textbf{sprue,batch}}$ sprue/runner surface area per batch [m^2] (from (PPI-5))

$t_{\textbf{shell}}$ total ceramic shell thickness [m]

$\rho_{\textbf{cer}}$ ceramic density [$\mathrm{kg/m}^3$]

$\phi_{\textbf{sol}}$ slurry solids fraction (mass fraction deposited) [–]

$\gamma_{\textbf{sprue}}$ sprue/runner area multiplier (dimensionless) [–]

Units note: The factor 10^{-9} converts $\mathrm{mm}^3 \to \mathrm{m}^3$ so that ρ_{fil} and ρ_{metal} (in $\mathrm{kg/m}^3$) produce kg before multiplying by $/kg prices.

Machine Cost per Part

Machine cost per part in printed pattern (Lost-PLA) investment casting includes:

- the cost of FFF printer usage, obtained by amortizing the printer's investment over its lifetime, utilization, and annual operating hours, and
- the cost of furnace usage, determined by the furnace hourly rate and distributed across the batch.

$$C^{\mathrm{mach}}_{/\mathrm{part}} = \frac{C^{/h}_{\mathrm{mach,ppi}}\, t_{\mathrm{build,ppi}}}{B_{\mathrm{batch}}} + \frac{C^{/h}_{\mathrm{furn}}\, t_{\mathrm{firing}}}{B_{\mathrm{batch}}} \tag{PPI-6}$$

$$C^{/h}_{\mathrm{mach,ppi}} = \frac{C_{\mathrm{inv,ppi}}}{\eta_{\mathrm{mach,ppi}}\, H_{\mathrm{year,ppi}}} \tag{PPI-7}$$

$$C_{\mathrm{inv,ppi}} = \frac{C_{\mathrm{mach,ppi}}}{L_{\mathrm{mach,ppi}}} \tag{PPI-8}$$

$$C^{/h}_{\mathrm{furn}} = \frac{C_{\mathrm{inv,furn}}}{\eta_{\mathrm{furn}}\, H_{\mathrm{year,furn}}} \tag{PPI-9}$$

$$C_{\mathrm{inv,furn}} = \frac{C_{\mathrm{mach,furn}}}{L_{\mathrm{mach,furn}}} \tag{PPI-10}$$

where:

$C^{\mathbf{mach}}_{\mathbf{/part}}$ machine cost per part [$]

$C^{/h}_{\mathbf{mach,ppi}}$ FFF printer hourly cost [$/h]

$t_{\mathbf{build,ppi}}$ FFF pattern build time [h]

$B_{\mathbf{batch}}$ batch size [–]

$C^{/h}_{\mathbf{furn}}$ furnace hourly rate [$/h]

$t_{\mathbf{firing}}$ furnace firing time [h]

$C_{\mathbf{inv,ppi}}$ yearly investment cost (printer) [$/year]

$\eta_{\mathbf{mach,ppi}}$ printer utilization [–]

$H_{\mathbf{year,ppi}}$ Total printer hours per year [h/year]

$C_{\mathbf{mach,ppi}}$ printer acquisition cost [$]

$L_{\mathbf{mach,ppi}}$ printer lifetime [years]

$C_{\mathbf{inv,furn}}$ annualized furnace investment cost [$/year]

$C_{\mathbf{mach,furn}}$ furnace acquisition cost [$]

$L_{\mathbf{mach,furn}}$ furnace lifetime [years]

$\eta_{\mathbf{furn}}$ furnace utilization [–]

$H_{\mathbf{year,furn}}$ Total furnace hours per year [h/year]

Labor Cost per Part

Labor cost per part in printed pattern (Lost-PLA) investment casting is divided into two contributions:

- **FDM pattern stage:** combined man-hours for printer setup and pattern postprocessing, and
- **Investment casting stage:** combined man-hours for cooling, injection, pattern/cluster assembly, coating, cutoff, breakout, and blast cleaning.

$$C^{\mathrm{lab}}_{/\mathrm{part}} = r_{\mathrm{lab}} \left(t_{\mathrm{fdm},/\mathrm{part}} + t_{\mathrm{cast},/\mathrm{part}}\right) + C^{/\mathrm{part}}_{\mathrm{breakout}} + C^{/\mathrm{part}}_{\mathrm{blast}} \tag{PPI-11}$$

$$t_{\mathrm{fdm},/\mathrm{part}} = \frac{t_{\mathrm{fdm},/\mathrm{batch}}}{B_{\mathrm{batch}}} \tag{PPI-12}$$

$$t_{\mathrm{cast},/\mathrm{part}} = \frac{t_{\mathrm{cast},/\mathrm{batch}}}{B_{\mathrm{batch}}} \tag{PPI-13}$$

where:

$C^{\mathbf{lab}}_{\mathbf{/part}}$ labor cost per part [$]

$r_{\mathbf{lab}}$ labor rate [$/h]

$t_{\mathbf{fdm,/part}}$ per-part man-hours for the FDM pattern stage [h/part]

$t_{\mathbf{cast,/part}}$ per-part man-hours for the investment casting stage [h/part]

$t_{\mathbf{fdm,/batch}}$ total FDM man-hours for the batch/cluster [h]

$t_{\mathbf{cast,/batch}}$ total casting man-hours for the batch/cluster [h]

$B_{\mathbf{batch}}$ batch size [–]

$C^{\mathbf{/part}}_{\mathbf{breakout}}$ breakout cost per part [$]

$C^{\mathbf{/part}}_{\mathbf{blast}}$ blast cleaning cost per part [$]

Energy Cost per Part

Energy cost per part in printed pattern (Lost-PLA) investment casting includes:

- the printer's electrical consumption during the FFF build (allocated over the batch), and
- the foundry furnace energy for melting the casting metal (per part, adjusted for furnace efficiency).

$$C^{\text{eng}}_{/\text{part}} = \frac{r_{\text{elec}}\, P_{\text{print,ppi}}\, t_{\text{build,ppi}}}{B_{\text{batch}}} + C_{\text{melt}} \tag{PPI-14}$$

$$C_{\text{melt}} = r_{\text{elec}}\, \frac{E_{\text{melt}}}{\eta_{\text{furn}}}\, f_{\text{melt}}\, \rho_{\text{metal}}\, V_{\text{part}} \times 10^{-9} \tag{PPI-15}$$

where:

$C^{\mathbf{eng}}_{\mathbf{/part}}$ energy cost per part [$]

$r_{\mathbf{elec}}$ electricity rate [$/kWh]

$P_{\mathbf{print,ppi}}$ average power draw of the FFF printer [kW]

$t_{\mathbf{build,ppi}}$ FFF pattern build time [h]

$B_{\mathbf{batch}}$ batch size [–]

$C_{\mathbf{melt}}$ melting energy cost per part [$]

$E_{\mathbf{melt}}$ specific energy to melt casting metal [kWh/kg]

$\eta_{\mathbf{furn}}$ furnace efficiency (0–1) [–]

$f_{\mathbf{melt}}$ melt allowance factor (e.g., 1.2) [–]

$\rho_{\mathbf{metal}}$ casting metal density [kg/m^3]

$V_{\mathbf{part}}$ net part volume [mm^3]

Note: The total melted mass per batch is $f_{\text{melt}}\, B_{\text{batch}}\, \rho_{\text{metal}}\, V_{\text{part}} \times 10^{-9}$. Thus, B_{batch} cancels in the melt term, making C_{melt} independent of batch size for identical parts; only the printer term scales with $1/B_{\text{batch}}$.

Post-Processing Cost per Part

Post-processing cost per part in printed pattern (Lost-PLA) investment casting is modeled as:

- a percentage factor representing finishing operations (e.g., surface finishing, minor touch-ups),
- applied to the direct cost per part excluding labor (material + machine + energy).

$$C^{\text{post}}_{/\text{part}} = f_{\text{post}}\, C^{\text{direct,nolab}}_{/\text{part}} \tag{PPI-16}$$

$$C^{\text{direct,nolab}}_{/\text{part}} = C^{\text{mat}}_{/\text{part}} + C^{\text{mach}}_{/\text{part}} + C^{\text{eng}}_{/\text{part}} \tag{PPI-17}$$

where:

$C^{\textbf{post}}_{\textbf{/part}}$ post-processing cost per part [$/part]

$f_{\textbf{post}}$ post-processing factor (e.g., 0.05–0.20) [–]

$C^{\textbf{direct,nolab}}_{\textbf{/part}}$ direct cost per part excluding labor [$/part]

$C^{\textbf{mat}}_{\textbf{/part}}$ material cost per part [$/part]

$C^{\textbf{mach}}_{\textbf{/part}}$ machine cost per part [$/part]

$C^{\textbf{eng}}_{\textbf{/part}}$ energy cost per part [$/part]

Maintenance Cost per Part

Maintenance cost per part in printed pattern (Lost-PLA) investment casting is modeled as:

- a yearly maintenance expense, expressed as a percentage of the machine acquisition cost, and
- an allocation of this yearly expense across the available machine hours and the number of parts produced.

$$C_{\mathrm{mnt}}^{/year} = \alpha_{\mathrm{mnt}}\, C_{\mathrm{mach,ppi}} \tag{PPI-18}$$

$$C_{/\mathrm{part}}^{\mathrm{mnt}} = \frac{C_{\mathrm{mnt}}^{/year}}{\eta_{\mathrm{mach,ppi}}\, H_{\mathrm{year,ppi}}}\, \frac{t_{\mathrm{build,ppi}}}{B_{\mathrm{batch}}} \tag{PPI-19}$$

where:

$C_{\mathbf{mnt}}^{/year}$ yearly maintenance cost [$/year]

$\alpha_{\mathbf{mnt}}$ maintenance fraction of acquisition (per year) [–]

$C_{\mathbf{mach,ppi}}$ FFF printer acquisition cost [$]

$C_{\mathbf{/part}}^{\mathbf{mnt}}$ maintenance cost per part [$/part]

$\eta_{\mathbf{mach,ppi}}$ printer utilization [–]

$H_{\mathbf{year,ppi}}$ total printer hours per year [h/year]

$t_{\mathbf{build,ppi}}$ FFF pattern build time [h]

$B_{\mathbf{batch}}$ batch size [–]

Overhead Cost per Part

Overhead cost per part in printed pattern (Lost-PLA) investment casting is modeled as:

- a yearly overhead expense, expressed as a percentage of the machine acquisition cost, and
- an allocation of this yearly overhead across the available machine hours and the number of parts produced.

$$C^{/year}_{\text{oh,ppi}} = \beta_{\text{oh,ppi}}\, C_{\text{mach,ppi}} \tag{PPI-20}$$

$$C^{/h}_{\text{oh,ppi}} = \frac{C^{/year}_{\text{oh,ppi}}}{\eta_{\text{mach,ppi}}\, H_{\text{year,ppi}}} \tag{PPI-21}$$

$$C^{\text{oh}}_{/\text{part}} = \frac{C^{/h}_{\text{oh,ppi}}\, t_{\text{build,ppi}}}{B_{\text{batch}}} \tag{PPI-22}$$

where:

$C^{/year}_{\mathbf{oh,ppi}}$ annual overhead cost [$/year]

$\beta_{\mathbf{oh,ppi}}$ overhead percentage of acquisition [–]

$C_{\mathbf{mach,ppi}}$ printer acquisition cost [$]

$C^{/h}_{\mathbf{oh,ppi}}$ overhead rate [$/h]

$\eta_{\mathbf{mach,ppi}}$ printer utilization [–]

$H_{\mathbf{year,ppi}}$ total printer hours per year [h/year]

$t_{\mathbf{build,ppi}}$ FFF pattern build time [h]

$B_{\mathbf{batch}}$ batch size [–]

$C^{\mathbf{oh}}_{\mathbf{/part}}$ overhead cost per part [$/part]

Total Cost per Part

The total cost per part in printed pattern (Lost-PLA) investment casting is the sum of:

- material cost (filament, slurry, and casting metal),
- machine cost (pattern printer and furnace),
- labor cost (FFF and casting operations),
- energy cost (FFF printing and metal melting),
- post-processing cost,
- maintenance cost, and
- overhead cost.

$$C_{/\text{part}}^{\text{total}} = C_{/\text{part}}^{\text{mat}} + C_{/\text{part}}^{\text{mach}} + C_{/\text{part}}^{\text{lab}} + C_{/\text{part}}^{\text{eng}} + C_{/\text{part}}^{\text{post}} + C_{/\text{part}}^{\text{mnt}} + C_{/\text{part}}^{\text{oh}} \tag{PPI-23}$$

where:

$C_{\mathbf{/part}}^{\mathbf{total}}$ total cost per part [$/part]

$C_{\mathbf{/part}}^{\mathbf{mat}}$ material cost per part [$/part]

$C_{\mathbf{/part}}^{\mathbf{mach}}$ machine cost per part [$/part]

$C_{\mathbf{/part}}^{\mathbf{lab}}$ labor cost per part [$/part]

$C_{\mathbf{/part}}^{\mathbf{eng}}$ energy cost per part [$/part]

$C_{\mathbf{/part}}^{\mathbf{post}}$ post-processing cost per part [$/part]

$C_{\mathbf{/part}}^{\mathbf{mnt}}$ maintenance cost per part [$/part]

$C_{\mathbf{/part}}^{\mathbf{oh}}$ overhead cost per part [$/part]

Laser Powder Bed Fusion (LPBF)

Material Cost per Part

Material cost per part in LPBF includes:

- the cost of powder consumed, scaled by a recycling factor ($f_{\text{recycle}} = 1/\eta_{\text{powder}}$), and
- the additional support volume, expressed as a support factor ($f_{\text{support}} = 1 + V_{\text{support}}/V_{\text{part}}$).

$$C^{\text{mat}}_{/\text{part}} = r_{\text{mat}}\ \rho_{\text{metal}} \left(\frac{V_{\text{part}}}{10^9}\right) \left(f_{\text{support}}\ f_{\text{recycle}}\right) \quad \text{(LPBF-1)}$$

$$f_{\text{recycle}} = \frac{1}{\eta_{\text{powder}}} \quad \text{(LPBF-2)}$$

$$f_{\text{support}} = 1 + \frac{V_{\text{support}}}{V_{\text{part}}} \quad \text{(LPBF-3)}$$

where:

$\boldsymbol{C^{\text{mat}}_{/\text{part}}}$ material cost per part [$]

$\boldsymbol{r_{\text{mat}}}$ material cost per unit mass [$/kg]

$\boldsymbol{\rho_{\text{metal}}}$ material density [kg/m^3]

$\boldsymbol{V_{\text{part}}}$ volume of part [mm^3] (divide by 10^9 to convert to m^3)

$\boldsymbol{V_{\text{support}}}$ volume of support material [mm^3]

$\boldsymbol{f_{\text{support}}}$ support factor [–]

$\boldsymbol{\eta_{\text{powder}}}$ powder efficiency (0–1) [–]

$\boldsymbol{f_{\text{recycle}}}$ recycling factor $= 1/\eta_{\text{powder}}$ [–]

Machine Cost per Part

Machine cost per part in LPBF includes:

- the setup cost allocated across the batch, and
- the machine usage cost, based on amortized investment per hour multiplied by build time and divided by the batch size.

$$C^{\text{mach}}_{/\text{part}} = \frac{C_{\text{setup,lpbf}}}{B_{\text{batch}}} + \frac{C^{/h}_{\text{mach,lpbf}}\, t_{\text{build,lpbf}}}{B_{\text{batch}}} \tag{LPBF-4}$$

$$C^{/h}_{\text{mach,lpbf}} = \frac{C_{\text{inv,lpbf}}}{\eta_{\text{mach,lpbf}}\, H_{\text{year,lpbf}}} \tag{LPBF-5}$$

$$C_{\text{inv,lpbf}} = \frac{C_{\text{mach,lpbf}}}{L_{\text{mach,lpbf}}} \tag{LPBF-6}$$

where:

$C^{\mathbf{mach}}_{\mathbf{/part}}$ machine cost per part [$/part]

$C_{\mathbf{setup,lpbf}}$ setup cost [$]

$B_{\mathbf{batch}}$ batch size [–]

$C^{/h}_{\mathbf{mach,lpbf}}$ machine hourly cost [$/h]

$t_{\mathbf{build,lpbf}}$ build time [h]

$C_{\mathbf{inv,lpbf}}$ annualized investment cost [$/year]

$\eta_{\mathbf{mach,lpbf}}$ machine utilization [–]

$H_{\mathbf{year,lpbf}}$ total available machine hours per year [h/year]

$C_{\mathbf{mach,lpbf}}$ machine acquisition cost [$]

$L_{\mathbf{mach,lpbf}}$ useful life of machine [years]

Labor Cost per Part

Labor cost per part in LPBF is modeled as:

- the total labor hours required for all activities during a build, multiplied by the labor rate, and
- distributed across the number of parts in the build.

$$C^{\text{lab}}_{/\text{part}} = \frac{r_{\text{lab}}\, t_{\text{task}}}{B_{\text{batch}}} \tag{LPBF-7}$$

where:

$C^{\mathbf{lab}}_{\mathbf{/part}}$ labor cost per part [$/part]

$r_{\mathbf{lab}}$ labor rate [$/h]

$t_{\mathbf{task}}$ total labor time for all tasks [h]

$B_{\mathbf{batch}}$ batch size [–]

Energy Cost per Part

Energy cost per part in LPBF includes:

- the electrical energy consumed by the LPBF machine during the build,
- allocated across all parts produced in the build.

$$C^{\mathrm{eng}}_{/\mathrm{part}} = \frac{r_{\mathrm{elec}}\, P_{\mathrm{avg,lpbf}}\, t_{\mathrm{build,lpbf}}}{B_{\mathrm{batch}}} \qquad \text{(LPBF-8)}$$

where:

$C^{\mathbf{eng}}_{\mathbf{/part}}$ energy cost per part [$/part]

$r_{\mathbf{elec}}$ electricity rate [$/kWh]

$P_{\mathbf{avg,lpbf}}$ average machine power [kW]

$t_{\mathbf{build,lpbf}}$ build time [h]

$B_{\mathbf{batch}}$ batch size [–]

Post-Processing Cost per Part

Post-processing cost per part in LPBF is modeled as:

- a percentage factor representing finishing operations (e.g., support removal, surface finishing),
- applied to the direct cost per part excluding labor (material + machine + energy).

$$C^{\text{post}}_{/\text{part}} = f_{\text{post}}\, C^{\text{direct}\backslash\text{lab}}_{/\text{part}} \tag{LPBF-9}$$

$$C^{\text{direct}\backslash\text{lab}}_{/\text{part}} = C^{\text{mat}}_{/\text{part}} + C^{\text{mach}}_{/\text{part}} + C^{\text{eng}}_{/\text{part}} \tag{LPBF-10}$$

where:

$C^{\mathbf{post}}_{\mathbf{/part}}$ post-processing cost per part [$/part]

$f_{\mathbf{post}}$ post-processing factor [–]

$C^{\mathbf{direct\backslash lab}}_{\mathbf{/part}}$ direct cost per part excluding labor [$/part]

$C^{\mathbf{mat}}_{\mathbf{/part}}$ material cost per part [$/part]

$C^{\mathbf{mach}}_{\mathbf{/part}}$ machine cost per part [$/part]

$C^{\mathbf{eng}}_{\mathbf{/part}}$ energy cost per part [$/part]

Maintenance Cost per Part

Maintenance cost per part in LPBF is modeled as:

- a yearly maintenance expense, expressed as a percentage of the machine acquisition cost, and
- an allocation of this yearly expense across the available machine hours and the number of parts produced.

$$C_{\mathrm{mnt}}^{/year} = \alpha_{\mathrm{mnt}}\, C_{\mathrm{mach,lpbf}} \tag{LPBF-11}$$

$$C_{/\mathrm{part}}^{\mathrm{mnt}} = \frac{C_{\mathrm{mnt}}^{/year}}{\eta_{\mathrm{mach,lpbf}}\, H_{\mathrm{year,lpbf}}}\, \frac{t_{\mathrm{build,lpbf}}}{B_{\mathrm{batch}}} \tag{LPBF-12}$$

where:

$C_{\mathbf{mnt}}^{/year}$ yearly maintenance cost [$/year]

$\alpha_{\mathbf{mnt}}$ maintenance fraction of acquisition (e.g., 0.10 = 10%) [–]

$C_{\mathbf{mach,lpbf}}$ machine acquisition cost [$]

$C_{\mathbf{/part}}^{\mathbf{mnt}}$ maintenance cost per part [$/part]

$\eta_{\mathbf{mach,lpbf}}$ machine utilization [–]

$H_{\mathbf{year,lpbf}}$ total machine hours per year [h/year]

$t_{\mathbf{build,lpbf}}$ build time [h]

$B_{\mathbf{batch}}$ batch size [–]

Overhead Cost per Part

Overhead cost per part in LPBF is modeled as:

- a yearly overhead expense, expressed as a percentage of the machine acquisition cost, and
- an allocation of this yearly overhead across the machine's available hours and the number of parts produced.

$$C_{\text{oh,lpbf}}^{/year} = \beta_{\text{oh,lpbf}}\, C_{\text{mach,lpbf}} \tag{LPBF-13}$$

$$C_{\text{oh,lpbf}}^{/h} = \frac{C_{\text{oh,lpbf}}^{/year}}{\eta_{\text{mach,lpbf}}\, H_{\text{year,lpbf}}} \tag{LPBF-14}$$

$$C_{/\text{part}}^{\text{oh}} = \frac{C_{\text{oh,lpbf}}^{/h}\, t_{\text{build,lpbf}}}{B_{\text{batch}}} \tag{LPBF-15}$$

where:

$C_{\mathbf{oh,lpbf}}^{/year}$ annual overhead cost [$/year]

$\beta_{\mathbf{oh,lpbf}}$ overhead percentage of acquisition [–]

$C_{\mathbf{mach,lpbf}}$ machine acquisition cost [$]

$C_{\mathbf{oh,lpbf}}^{/h}$ overhead rate [$/h]

$\eta_{\mathbf{mach,lpbf}}$ machine utilization [–]

$H_{\mathbf{year,lpbf}}$ annual machine hours [h/year]

$t_{\mathbf{build,lpbf}}$ build time [h]

$B_{\mathbf{batch}}$ batch size [–]

$C_{\mathbf{/part}}^{\mathbf{oh}}$ overhead cost per part [$/part]

Total Cost per Part

The total cost per part in LPBF is the sum of:

- material cost (powder, including recycling and supports),
- machine cost (setup and amortized usage),
- labor cost,
- energy cost,
- post-processing cost,
- maintenance cost, and
- overhead cost.

$$C_{/\text{part}}^{\text{total}} = C_{/\text{part}}^{\text{mat}} + C_{/\text{part}}^{\text{mach}} + C_{/\text{part}}^{\text{lab}} + C_{/\text{part}}^{\text{eng}} + C_{/\text{part}}^{\text{post}} + C_{/\text{part}}^{\text{mnt}} + C_{/\text{part}}^{\text{oh}} \quad \text{(LPBF-16)}$$

where:

$C_{/\mathbf{part}}^{\mathbf{total}}$ total cost per part [$/part]

$C_{/\mathbf{part}}^{\mathbf{mat}}$ material cost per part [$/part]

$C_{/\mathbf{part}}^{\mathbf{mach}}$ machine cost per part [$/part]

$C_{/\mathbf{part}}^{\mathbf{lab}}$ labor cost per part [$/part]

$C_{/\mathbf{part}}^{\mathbf{eng}}$ energy cost per part [$/part]

$C_{/\mathbf{part}}^{\mathbf{post}}$ post-processing cost per part [$/part]

$C_{/\mathbf{part}}^{\mathbf{mnt}}$ maintenance cost per part [$/part]

$C_{/\mathbf{part}}^{\mathbf{oh}}$ overhead cost per part [$/part]

Machining

Material Cost per Part

Material cost per part in machining is modeled as:

- the mass of the *stock* required to produce the part (stock factor applied to net part volume),
- multiplied by the material unit price.

$$C^{\text{mat}}_{/\text{part}} = r_{\text{mat}}\,\rho_{\text{metal}}\left(V_{\text{stock}} \times 10^{-9}\right) \quad \text{(MACH-1)}$$

$$V_{\text{stock}} = f_{\text{stock}}\,V_{\text{part}} \quad \text{(MACH-2)}$$

where:

$C^{\mathbf{mat}}_{\mathbf{/part}}$ material cost per part [\$/part]

$r_{\mathbf{mat}}$ material unit price [\$/kg]

$\rho_{\mathbf{metal}}$ material density [kg/m^3]

$V_{\mathbf{stock}}$ stock volume for one part [mm^3]

$V_{\mathbf{part}}$ net part volume [mm^3]

$f_{\mathbf{stock}}$ stock factor (buy-to-make ratio), $f_{\text{stock}} \geq 1$ [–]

Machine Cost per Part

Machine cost per part in machining is composed of:

- the amortized machine investment cost per hour, based on capital cost, installation, miscellaneous cost, and equipment life, and
- the machining time required to remove the necessary material volume, multiplied by the machine hourly rate and allocated per part.

$$C^{\text{mach}}_{/\text{part}} = \frac{C^{/h}_{\text{mach,mach}}\ t_{\text{mach}}}{B_{\text{batch}}} \tag{MACH-3}$$

$$C^{/h}_{\text{mach,mach}} = \frac{C_{\text{inv}}}{\eta_{\text{mach,mach}}\ H_{\text{year,mach}}} \tag{MACH-4}$$

$$C_{\text{inv}} = \frac{C_{\text{capital}}\ (1 + f_{\text{install}}) + C_{\text{misc}}}{L_{\text{equip}}} \tag{MACH-5}$$

$$t_{\text{mach}} = \frac{V_{\text{rem}}}{\dot{V}_{\text{MRR}}} \times \frac{1}{60} \tag{MACH-6}$$

where:

$C^{\text{mach}}_{/\text{part}}$ machine cost per part [$/part]

$C^{/h}_{\text{mach,mach}}$ machine hourly cost [$/h]

t_{mach} machining time [h]

B_{batch} batch size [–]

C_{inv} yearly investment cost [$/year]

$\eta_{\text{mach,mach}}$ machine utilization [–]

$H_{\text{year,mach}}$ total machine hours per year [h/year]

C_{capital} capital cost [$]

f_{install} installation cost fraction [–]

C_{misc} miscellaneous cost [$]

L_{equip} equipment life [years]

V_{rem} volume of material to be removed [mm^3]

$\dot{V}_{\text{MRR}}$ material removal rate [mm^3/min]

Labor Cost per Part

Labor cost per part in machining is modeled as:

- the labor time per part, which includes operator-attended machining time (scaled by the operator time fraction) and idle time, and
- multiplied by the labor rate and the number of operators.

$$C^{\text{lab}}_{/\text{part}} = r_{\text{lab}}\, N_{\text{ops}}\, t_{\text{lab,part}} \tag{MACH-7}$$

$$t_{\text{lab,part}} = f_{\text{op}}\, t_{\text{oper,part}} \;+\; \frac{t_{\text{idle,part}}}{60} \tag{MACH-8}$$

$$t_{\text{oper,part}} = \frac{V_{\text{rem,part}}}{\dot{V}_{\text{MRR}}} \;\times\; \frac{1}{60} \tag{MACH-9}$$

where:

$C^{\textbf{lab}}_{\textbf{/part}}$ labor cost per part [$/part]

$r_{\textbf{lab}}$ labor rate [$/h]

$N_{\textbf{ops}}$ number of operators [–]

$t_{\textbf{lab,part}}$ labor time per part [h]

$f_{\textbf{op}}$ operator time fraction [–]

$t_{\textbf{oper,part}}$ operating time per part [h]

$t_{\textbf{idle,part}}$ idle time per part [min/part]

$V_{\textbf{rem,part}}$ volume of material removed per part [mm^3]

$\dot{V}_{\textbf{MRR}}$ material removal rate [mm^3/min]

Energy Cost per Part

Energy cost per part in machining includes:

- the energy consumed during machining, proportional to the operating power and machining time, and
- the standby energy consumed during idle time.

$$C^{\text{eng}}_{/\text{part}} = r_{\text{elec}} \left(P_{\text{oper}}\, t_{\text{oper,part}} + P_{\text{standby}}\, t_{\text{idle,part}} \right) \tag{MACH-10}$$

$$t_{\text{oper,part}} = \frac{V_{\text{rem,part}}}{\dot{V}_{\text{MRR}}} \times \frac{1}{60} \tag{MACH-11}$$

where:

$C^{\mathbf{eng}}_{\mathbf{/part}}$ energy cost per part [$/part]

$r_{\mathbf{elec}}$ electricity rate [$/kWh]

$P_{\mathbf{oper}}$ operating power [kW]

$P_{\mathbf{standby}}$ standby power [kW]

$t_{\mathbf{oper,part}}$ machining operating time per part [h]

$t_{\mathbf{idle,part}}$ idle time per part [h]

$V_{\mathbf{rem,part}}$ volume to be removed per part [mm^3]

$\dot{V}_{\mathbf{MRR}}$ material removal rate [mm^3/min]

Tooling Cost per Part

Tooling cost per part in machining is modeled simply as:

- annualizing the tool/fixture purchase cost over its life, and
- allocating that annual cost over the *total* annual parts output (across all stations).

$$C^{\text{tool}}_{/\text{part}} = \frac{C_{\text{tool}}\, N_{\text{finish}}}{L_{\text{tool}}\, Q_{\text{prod}}} \tag{MACH-12}$$

where:

$C^{\textbf{tool}}_{\textbf{/part}}$ tooling cost per part [$/part]

$C_{\textbf{tool}}$ tooling/fixture cost for one station [$]

$N_{\textbf{finish}}$ number of identical stations/tool sets [–]

$L_{\textbf{tool}}$ tooling life [years]

$Q_{\textbf{prod}}$ annual production volume (total parts/year across all stations) [parts/year]

Maintenance Cost per Part

Maintenance cost per part in machining is modeled as:

- a yearly maintenance expense, expressed as a percentage of the machine acquisition cost, and
- allocation of this annual cost via an hourly rate across the per-part machining time.

$$C_{\text{mnt}}^{/year} = \alpha_{\text{mnt}}\, C_{\text{mach,mach}} \tag{MACH-13}$$

$$C_{\text{mnt}}^{/h} = \frac{C_{\text{mnt}}^{/year}}{\eta_{\text{mach,mach}}\, H_{\text{year,mach}}} \tag{MACH-14}$$

$$C_{/\text{part}}^{\text{mnt}} = C_{\text{mnt}}^{/h}\, t_{\text{mach}} \tag{MACH-15}$$

where:

$C_{\mathbf{mnt}}^{/year}$ yearly maintenance cost [$/year]

$\alpha_{\mathbf{mnt}}$ maintenance percentage of acquisition [–]

$C_{\mathbf{mach,mach}}$ machine acquisition cost [$]

$C_{\mathbf{mnt}}^{/h}$ maintenance rate [$/h]

$\eta_{\mathbf{mach,mach}}$ machine utilization [–]

$H_{\mathbf{year,mach}}$ total machine hours per year [h/year]

$t_{\mathbf{mach}}$ machining time per part [h]

$C_{\mathbf{/part}}^{\mathbf{mnt}}$ maintenance cost per part [$/part]

Overhead Cost per Part

Overhead cost per part in machining is modeled as:

- a yearly overhead expense, expressed as a percentage of the machine acquisition cost, and
- allocation of this annual overhead via an hourly rate across the per-part machining time.

$$C^{/year}_{\text{oh,mach}} = \beta_{\text{oh,mach}}\, C_{\text{mach,mach}} \tag{MACH-16}$$

$$C^{/h}_{\text{oh,mach}} = \frac{C^{/year}_{\text{oh,mach}}}{\eta_{\text{mach,mach}}\, H_{\text{year,mach}}} \tag{MACH-17}$$

$$C^{\text{oh}}_{/\text{part}} = C^{/h}_{\text{oh,mach}}\, t_{\text{mach}} \tag{MACH-18}$$

where:

$C^{/year}_{\textbf{oh,mach}}$ annual overhead cost (machining) [$/year]

$\beta_{\textbf{oh,mach}}$ overhead percentage of acquisition [–]

$C_{\textbf{mach,mach}}$ machine acquisition cost [$]

$C^{/h}_{\textbf{oh,mach}}$ overhead rate [$/h]

$\eta_{\textbf{mach,mach}}$ machine utilization [–]

$H_{\textbf{year,mach}}$ annual machine hours [h/year]

$t_{\textbf{mach}}$ machining time per part [h]

$C^{\textbf{oh}}_{\textbf{/part}}$ overhead cost per part [$/part]

Total Cost per Part

The total cost per part in machining is the sum of:

- material cost,
- machine cost,
- labor cost,
- energy cost,
- tooling cost,
- maintenance cost, and
- overhead cost.

$$
\begin{aligned}
C^{\text{total}}_{/\text{part}} = C^{\text{mat}}_{/\text{part}} + C^{\text{mach}}_{/\text{part}} + C^{\text{lab}}_{/\text{part}} + C^{\text{eng}}_{/\text{part}} \\
+ C^{\text{tool}}_{/\text{part}} + C^{\text{mnt}}_{/\text{part}} + C^{\text{oh}}_{/\text{part}}
\end{aligned} \tag{MACH-19}
$$

where:

$\boldsymbol{C^{\textbf{total}}_{\textbf{/part}}}$ total cost per part [$/part]

$\boldsymbol{C^{\textbf{mat}}_{\textbf{/part}}}$ material cost per part [$/part]

$\boldsymbol{C^{\textbf{mach}}_{\textbf{/part}}}$ machine cost per part [$/part]

$\boldsymbol{C^{\textbf{lab}}_{\textbf{/part}}}$ labor cost per part [$/part]

$\boldsymbol{C^{\textbf{eng}}_{\textbf{/part}}}$ energy cost per part [$/part]

$\boldsymbol{C^{\textbf{tool}}_{\textbf{/part}}}$ tooling cost per part [$/part]

$\boldsymbol{C^{\textbf{mnt}}_{\textbf{/part}}}$ maintenance cost per part [$/part]

$\boldsymbol{C^{\textbf{oh}}_{\textbf{/part}}}$ overhead cost per part [$/part]

Investment Casting

Material Cost per Part

Material cost per part in investment casting is modeled on a per-*cluster* basis:

- core and pattern material subtotals are computed for the entire cluster, then
- slurry (per coat per cluster) is scaled by the number of coats, and
- the casting metal is charged per part via a melt-allowance factor and a process-loss rate.

$$C^{\mathrm{mat}}_{/\mathrm{part}} = \frac{C_{\mathrm{core,cl}} + C_{\mathrm{pattern,cl}} + N_{\mathrm{coats}}\, C_{\mathrm{slurry,cl}}}{B_{\mathrm{batch}}} \;+\; C_{\mathrm{metal}} \tag{INV-1}$$

$$C_{\mathrm{core,cl}} = B_{\mathrm{batch}} \left(\rho_{\mathrm{core}}\, V_{\mathrm{core}}\right) r_{\mathrm{core}} \tag{INV-2}$$

$$C_{\mathrm{pattern,cl}} = B_{\mathrm{batch}} \left(\rho_{\mathrm{pattern}}\, V_{\mathrm{pattern}}\right) r_{\mathrm{pattern}} \tag{INV-3}$$

$$C_{\mathrm{slurry,cl}} = r_{\mathrm{slurry}}\, m_{\mathrm{slurry,cl}} \tag{INV-4}$$

$$m_{\mathrm{slurry,cl}} = \Big(A_{\mathrm{tree,part}}\, B_{\mathrm{batch}} + A_{\mathrm{sprue,cl}}\Big)\, t_{\mathrm{shell}}\, \rho_{\mathrm{cer}}\, \phi_{\mathrm{sol}} \tag{INV-5}$$

$$A_{\mathrm{sprue,cl}} = \gamma_{\mathrm{sprue}}\, B_{\mathrm{batch}}\, A_{\mathrm{tree,part}} \tag{INV-6}$$

$$C_{\mathrm{metal}} = \left(r_{\mathrm{mat}} + r_{\mathrm{loss}}\right) f_{\mathrm{melt}}\, m_{\mathrm{part}} \tag{INV-7}$$

$$m_{\mathrm{part}} = \rho_{\mathrm{metal}}\, V_{\mathrm{part}} \times 10^{-9} \tag{INV-8}$$

where:

$\boldsymbol{C^{\mathrm{mat}}_{/\mathrm{part}}}$ material cost per part [$/part]

$\boldsymbol{C_{\mathrm{core,cl}}}$ core material subtotal per cluster [$]

$\boldsymbol{\rho_{\mathrm{core}}}$ core density [kg/m^3]

$\boldsymbol{V_{\mathrm{core}}}$ core volume per part [m^3]

$\boldsymbol{r_{\mathrm{core}}}$ core material price [$/kg]

$\boldsymbol{C_{\mathrm{pattern,cl}}}$ pattern material subtotal per cluster [$]

$\boldsymbol{\rho_{\mathrm{pattern}}}$ pattern density [kg/m^3]

$\boldsymbol{V_{\mathrm{pattern}}}$ pattern volume per part [m^3]

$\boldsymbol{r_{\mathrm{pattern}}}$ pattern material price [$/kg]

$N_{\mathbf{coats}}$ number of coats [–]

$C_{\mathbf{slurry,cl}}$ slurry cost per cluster [$]

$r_{\mathbf{slurry}}$ slurry cost per unit mass [$/kg]

$m_{\mathbf{slurry,cl}}$ slurry mass per cluster [kg]

$A_{\mathbf{tree,part}}$ wax-tree surface area per part [m^2]

$A_{\mathbf{sprue,cl}}$ sprue/runner surface area per cluster [m^2]

$\gamma_{\mathbf{sprue}}$ sprue/runner area multiplier (dimensionless) [–]

$t_{\mathbf{shell}}$ total ceramic shell thickness [m]

$\rho_{\mathbf{cer}}$ ceramic density [$\mathrm{kg/m}^3$]

$\phi_{\mathbf{sol}}$ slurry solids fraction [–]

$B_{\mathbf{batch}}$ Batch size [–]

$C_{\mathbf{metal}}$ casting metal cost (per part) [$]

$r_{\mathbf{mat}}$ raw casting metal price [$/kg]

$r_{\mathbf{loss}}$ process-loss surcharge [$/kg]

$f_{\mathbf{melt}}$ melt allowance factor [–]

$m_{\mathbf{part}}$ mass of one finished part [kg]

$\rho_{\mathbf{metal}}$ casting metal density [$\mathrm{kg/m}^3$]

$V_{\mathbf{part}}$ net part volume [mm^3]

Machine Cost per Part

Machine cost per part in investment casting includes:

- furnace usage during shell firing/burnout, and
- allocation of that furnace cost across clusters in a batch and parts per cluster.

$$C^{\mathrm{mach}}_{/\mathrm{part}} = \frac{C^{/h}_{\mathrm{furn}}\, t_{\mathrm{firing}}}{N_{\mathrm{cl,batch}}\, B_{\mathrm{batch}}} \tag{INV-9}$$

$$C^{/h}_{\mathrm{furn}} = \frac{C_{\mathrm{inv,furn}}}{\eta_{\mathrm{furn}}\, H_{\mathrm{year,furn}}} \tag{INV-10}$$

$$C_{\mathrm{inv,furn}} = \frac{C_{\mathrm{mach,furn}}}{L_{\mathrm{mach,furn}}} \tag{INV-11}$$

where:

$C^{\mathbf{mach}}_{\mathbf{/part}}$ machine cost per part [$/part]

$C^{/h}_{\mathbf{furn}}$ furnace hourly rate [$/h]

$t_{\mathbf{firing}}$ furnace firing/burnout time [h]

$N_{\mathbf{cl,batch}}$ clusters processed per batch [–]

$B_{\mathbf{batch}}$ batch size [–]

$C_{\mathbf{inv,furn}}$ furnace investment cost [$/year]

$\eta_{\mathbf{furn}}$ furnace utilization [–]

$H_{\mathbf{year,furn}}$ total furnace hours per year [h/year]

$C_{\mathbf{mach,furn}}$ furnace acquisition cost [$]

$L_{\mathbf{mach,furn}}$ furnace lifetime [years]

Labor Cost per Part

Labor cost per part in investment casting is modeled as:

- combined man-hours for the investment casting stage (cooling, injection, pattern/cluster assembly, coating, special operations, cutoff), allocated per part, and
- breakout and blast cleaning costs per part.

$$C^{\text{lab}}_{/\text{part}} = r_{\text{lab}}\, t_{\text{cast},/\text{part}} + C^{/\text{part}}_{\text{breakout}} + C^{/\text{part}}_{\text{blast}} \tag{INV-12}$$

$$t_{\text{cast},/\text{part}} = \frac{t_{\text{cast},/\text{batch}}}{B_{\text{batch}}} \tag{INV-13}$$

where:

$C^{\textbf{lab}}_{\textbf{/part}}$ labor cost per part [$/part]

$r_{\textbf{lab}}$ labor rate [$/h]

$t_{\textbf{cast,/part}}$ per-part man-hours for the investment casting stage [h/part]

$t_{\textbf{cast,/batch}}$ total casting-stage man-hours for the batch/cluster [h]

$B_{\textbf{batch}}$ batch size [–]

$C^{\textbf{/part}}_{\textbf{breakout}}$ breakout cost per part [$/part]

$C^{\textbf{/part}}_{\textbf{blast}}$ blast cleaning cost per part [$/part]

Energy Cost per Part

Energy cost per part in investment casting includes:

- the furnace energy required to melt the casting metal, allocated per part, and
- adjustment for furnace efficiency.

$$C^{\text{eng}}_{/\text{part}} = r_{\text{elec}} \, \frac{E^{/kg}_{\text{melt}}}{\eta_{\text{furn}}} \, m_{\text{metal,part}} \tag{INV-14}$$

$$m_{\text{metal,part}} = f_{\text{melt}} \, \rho_{\text{metal}} \, V_{\text{part}} \times 10^{-9} \tag{INV-15}$$

where:

$C^{\textbf{eng}}_{\textbf{/part}}$ energy cost per part [$/part]

$r_{\textbf{elec}}$ electricity rate [$/kWh]

$E^{/kg}_{\textbf{melt}}$ specific energy required to melt metal [kWh/kg]

$\eta_{\textbf{furn}}$ furnace efficiency [–]

$m_{\textbf{metal,part}}$ mass of metal cast per part [kg/part]

$f_{\textbf{melt}}$ melt allowance factor (e.g., 1.2) [–]

$\rho_{\textbf{metal}}$ casting metal density [kg/m^3]

$V_{\textbf{part}}$ net part volume [mm^3]

Tooling Cost per Part

Tooling cost per part in investment casting is annualized and allocated across total annual output:

- annualize the purchase cost of the tooling/fixtures over their service life, and
- divide by the *total* annual production volume (parts/year across all clusters and stations).

$$C^{\text{tool}}_{/\text{part}} = \frac{C_{\text{tool}}\, N_{\text{sets}}}{L_{\text{tool}}\, Q_{\text{prod}}} \tag{INV-16}$$

where:

$\boldsymbol{C^{\text{tool}}_{/\text{part}}}$ tooling cost per part [$/part]

$\boldsymbol{C_{\text{tool}}}$ tooling/fixture purchase cost for one set (e.g., wax die, core box) [$]

$\boldsymbol{N_{\text{sets}}}$ number of identical tool sets in service (parallel stations) [–]

$\boldsymbol{L_{\text{tool}}}$ tooling service life [years]

$\boldsymbol{Q_{\text{prod}}}$ annual production volume (total finished parts/year across all stations) [parts/year]

Total Cost per Part

The total cost per part in investment casting is the sum of:

- material cost,
- machine cost,
- labor cost,
- energy cost, and
- tooling cost.

$$C^{\text{total}}_{/\text{part}} = C^{\text{mat}}_{/\text{part}} + C^{\text{mach}}_{/\text{part}} + C^{\text{lab}}_{/\text{part}} + C^{\text{eng}}_{/\text{part}} + C^{\text{tool}}_{/\text{part}} \tag{INV-17}$$

where:

$\boldsymbol{C^{\textbf{total}}_{\textbf{/part}}}$ total cost per part [$/part]

$\boldsymbol{C^{\textbf{mat}}_{\textbf{/part}}}$ material cost per part [$/part]

$\boldsymbol{C^{\textbf{mach}}_{\textbf{/part}}}$ machine cost per part [$/part]

$\boldsymbol{C^{\textbf{lab}}_{\textbf{/part}}}$ labor cost per part [$/part]

$\boldsymbol{C^{\textbf{eng}}_{\textbf{/part}}}$ energy cost per part [$/part]

$\boldsymbol{C^{\textbf{tool}}_{\textbf{/part}}}$ tooling cost per part [$/part]

Sand Casting

Material Cost per Part

Material cost per part in sand casting is modeled as:

- **direct metal cost per part** (casting weight × unit metal price × loss factor), and
- **indirect sand costs per mold** (mold sand + core sand) allocated over the number of parts per mold.

$$C^{\text{mat}}_{/\text{part}} = C^{\text{direct}}_{/\text{part}} + \frac{C_{\text{indirect}}}{B_{\text{batch}}} \tag{SAND-1}$$

$$C^{\text{direct}}_{/\text{part}} = r_{\text{metal}}\, m_{\text{cast}}\, f_{\text{loss}} \tag{SAND-2}$$

$$m_{\text{cast}} = \rho_{\text{metal}}\, V_{\text{cast}} \tag{SAND-3}$$

$$C_{\text{indirect}} = C_{\text{sand}} + C_{\text{core}} \tag{SAND-4}$$

$$C_{\text{sand}} = r_{\text{sand}}\, f_{\text{sand,recycle}}\, f_{\text{reject,cast-mold-sand}}\, \rho_{\text{sand}}\, V_{\text{sand}} \tag{SAND-5}$$

$$C_{\text{core}} = r_{\text{core}}\, f_{\text{sand,recycle}}\, f_{\text{reject,cast-mold-sand}}\, \rho_{\text{core,sand}}\, V_{\text{core,sand}} \tag{SAND-6}$$

$$V_{\text{sand}} = k_{\text{box}}\, 1.5\, B_{\text{batch}}\, V_{\text{box,part}} \tag{SAND-7}$$

where:

$C^{\mathbf{mat}}_{\mathbf{/part}}$ material cost per part [$/part]

$C^{\mathbf{direct}}_{\mathbf{/part}}$ direct metal cost per part [$/part]

$r_{\mathbf{metal}}$ unit metal cost [$/kg]

$m_{\mathbf{cast}}$ casting weight (net metal per part) [kg]

$\rho_{\mathbf{metal}}$ casting metal density [kg/m^3]

$V_{\mathbf{cast}}$ casting volume [m^3]

$f_{\mathbf{loss}}$ factor for metal loss [–]

$C_{\mathbf{indirect}}$ indirect sand-related cost per mold [$]

$C_{\mathbf{sand}}$ mold sand cost per mold [$]

$r_{\mathbf{sand}}$ price of mold sand mixture [$/kg]

$f_{\mathbf{sand,recycle}}$ factor for recycled sand [–]

$f_{\textbf{reject,cast-mold-sand}}$ factor for casting, mold, and sand rejection [–]

$\rho_{\textbf{sand}}$ density of mold sand [kg/m^3]

$V_{\textbf{sand}}$ total volume of mold sand per batch [m^3] (see (SAND-7))

$C_{\textbf{core}}$ core sand cost per mold [\$]

$r_{\textbf{core}}$ price of core sand mixture [\$/kg]

$\rho_{\textbf{core,sand}}$ density of core sand [kg/m^3]

$V_{\textbf{core,sand}}$ total volume of core sand [m^3]

$B_{\textbf{batch}}$ batch size [–]

$k_{\textbf{box}}$ mold bounding-box factor (accounts for flask, riser, gating, etc.) [–]

$V_{\textbf{box,part}}$ bounding-box volume of one part [m^3]

Machine Cost per Part

Machine cost per part in sand casting captures the furnace usage during the melt/hold/pour window, allocated across the parts represented in that furnace batch.

$$C^{\mathrm{mach}}_{/\mathrm{part}} = \frac{C^{/h}_{\mathrm{furn}}\ t_{\mathrm{batch}}}{B_{\mathrm{batch}}} \tag{SAND-8}$$

$$C^{/h}_{\mathrm{furn}} = \frac{C_{\mathrm{inv,furn}}}{\eta_{\mathrm{furn}}\ H_{\mathrm{year,furn}}} \tag{SAND-9}$$

$$C_{\mathrm{inv,furn}} = \frac{C_{\mathrm{mach,furn}}}{L_{\mathrm{mach,furn}}} \tag{SAND-10}$$

where:

$C^{\mathbf{mach}}_{\mathbf{/part}}$ machine cost per part [$/part]

$C^{/h}_{\mathbf{furn}}$ furnace hourly rate [$/h]

$t_{\mathbf{batch}}$ furnace time for the melt/hold/pour window [h]

$B_{\mathbf{batch}}$ batch size [–]

$C_{\mathbf{inv,furn}}$ furnace investment cost per year [$/year]

$\eta_{\mathbf{furn}}$ furnace utilization [–]

$H_{\mathbf{year,furn}}$ available furnace hours per year [h/year]

$C_{\mathbf{mach,furn}}$ furnace acquisition cost [$]

$L_{\mathbf{mach,furn}}$ furnace lifetime [years]

Labor Cost per Part

Labor cost per part in sand casting is modeled as:

- technician man-hours multiplied by the hourly rate,
- adjusted by casting and activity rejection factors, and
- allocated across the batch size.

$$C^{\text{lab}}_{\text{/part}} = \frac{f_{\text{reject,cast}}\, f_{\text{reject,act}}\, r_{\text{tech}}\, t_{\text{man}}}{B_{\text{batch}}} \tag{SAND-11}$$

where:

$C^{\mathbf{lab}}_{\mathbf{/part}}$ labor cost per part [$/part]

$f_{\mathbf{reject,cast}}$ factor for casting rejection [–]

$f_{\mathbf{reject,act}}$ activity rejection factor [–]

$r_{\mathbf{tech}}$ technician hourly rate [$/h]

$t_{\mathbf{man}}$ technician man-hours for the activity [h]

$B_{\mathbf{batch}}$ batch size [–]

Energy Cost per Part

Energy cost per part in sand casting includes the furnace energy to **melt** the metal and to **hold** it at temperature, adjusted for furnace efficiency. For identical parts, the per-part form is:

$$C^{\mathrm{eng}}_{/\mathrm{part}} = \frac{r_{\mathrm{elec}}}{\eta_{\mathrm{furn}}} \left[E^{/kg}_{\mathrm{melt}} \; + \; E^{/(kg\cdot h)}_{\mathrm{hold}} \, t_{\mathrm{hold}} \right] f_{\mathrm{melt}} \, \rho_{\mathrm{metal}} \, V_{\mathrm{part}} \times 10^{-9} \qquad \text{(SAND-12)}$$

where:

$C^{\mathbf{eng}}_{\mathbf{/part}}$ energy cost per part [$/part]

$r_{\mathbf{elec}}$ electricity rate [$/kWh]

$E^{/kg}_{\mathbf{melt}}$ specific energy to melt metal [kWh/kg]

$E^{/(kg\cdot h)}_{\mathbf{hold}}$ specific energy to hold melt temperature [kWh/(kg·h)]

$t_{\mathbf{hold}}$ holding time [h]

$\eta_{\mathbf{furn}}$ furnace efficiency [–]

$f_{\mathbf{melt}}$ melt allowance factor (e.g., 1.2) [–]

$\rho_{\mathbf{metal}}$ casting metal density [kg/m^3]

$V_{\mathbf{part}}$ net part volume [mm^3]

Tooling Cost per Part

Tooling cost per part in sand casting is annualized and allocated across total annual output:

- annualize the purchase cost of tooling/fixtures over their service life, and
- divide by the *total* annual production volume (finished parts/year across all lines).

$$C^{\text{tool}}_{/\text{part}} = \frac{C_{\text{tool}}\, N_{\text{sets}}}{L_{\text{tool}}\, Q_{\text{prod}}} \tag{SAND-13}$$

where:

$\boldsymbol{C^{\text{tool}}_{/\text{part}}}$ tooling cost per part [$/part]

$\boldsymbol{C_{\text{tool}}}$ tooling/fixture purchase cost for one set [$]

$\boldsymbol{N_{\text{sets}}}$ number of identical tool sets in service (parallel molds/lines) [–]

$\boldsymbol{L_{\text{tool}}}$ tooling service life [years]

$\boldsymbol{Q_{\text{prod}}}$ annual production volume (total finished parts/year across all lines) [parts/year]

Overhead Cost per Part

Overhead cost per part in sand casting is modeled as:

- the annual overhead cost of foundry operations, expressed as a fraction of machine acquisition cost (per year),
- distributed over yearly available machine hours (adjusted by utilization), and
- allocated to each part based on batch time and the batch size.

$$C_{\text{oh,sand}}^{/year} = \beta_{\text{oh,sand}}\, C_{\text{mach,sand}} \tag{SAND-14}$$

$$C_{\text{oh,sand}}^{/h} = \frac{C_{\text{oh,sand}}^{/year}}{\eta_{\text{mach,sand}}\, H_{\text{year,sand}}} \tag{SAND-15}$$

$$C_{/\text{part}}^{\text{oh,sand}} = \frac{C_{\text{oh,sand}}^{/h}\, t_{\text{batch}}}{B_{\text{batch}}} \tag{SAND-16}$$

where:

$C_{\mathbf{/part}}^{\mathbf{oh,sand}}$ overhead cost per part [$/part]

$C_{\mathbf{oh,sand}}^{/h}$ overhead rate [$/h]

$C_{\mathbf{oh,sand}}^{/year}$ annual overhead cost [$/year]

$\beta_{\mathbf{oh,sand}}$ overhead fraction of acquisition (per year) [–]

$C_{\mathbf{mach,sand}}$ machine acquisition cost (sand casting) [$]

$\eta_{\mathbf{mach,sand}}$ machine utilization (sand casting) [–]

$H_{\mathbf{year,sand}}$ annual machine hours (sand casting) [h/year]

$t_{\mathbf{batch}}$ batch time [h]

$B_{\mathbf{batch}}$ batch size [–]

Total Cost per Part

The total cost per part in sand casting is the sum of:

- material cost,
- machine cost,
- labor cost,
- energy cost,
- tooling cost, and
- overhead cost.

$$C^{\text{total}}_{/\text{part}} = C^{\text{mat}}_{/\text{part}} + C^{\text{mach}}_{/\text{part}} + C^{\text{lab}}_{/\text{part}} + C^{\text{eng}}_{/\text{part}} + C^{\text{tool}}_{/\text{part}} + C^{\text{oh,sand}}_{/\text{part}} \quad \text{(SAND-17)}$$

where:

$C^{\textbf{total}}_{\textbf{/part}}$ total cost per part [$/part]

$C^{\textbf{mat}}_{\textbf{/part}}$ material cost per part [$/part]

$C^{\textbf{mach}}_{\textbf{/part}}$ machine cost per part [$/part]

$C^{\textbf{lab}}_{\textbf{/part}}$ labor cost per part [$/part]

$C^{\textbf{eng}}_{\textbf{/part}}$ energy cost per part [$/part]

$C^{\textbf{tool}}_{\textbf{/part}}$ tooling cost per part [$/part]

$C^{\textbf{oh,sand}}_{\textbf{/part}}$ overhead cost per part [$/part]